Data Mining
Homework 2

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1. Question 1

   Explain the following normalization methods based on their formula

   a.) Min-Max normalization

   b.) Z-score normalization

   c.) Apply normalization to the data vector $X = [200, 300, 400, 600, 1000]$.

      i.) Use min-max normalization with min $= 0$ and max $= 1$.

      ii.) z-score normalization

   d.) Append a new data point of 250. Use the same z-score transformation from before to transform the data.

Ans:

a.) The Min-Max Normalization for data points $x_i \in X$ rescaled into $X^* \in [0, 1]$ is given by,

$$x_i^* = \frac{x_i - \min(X)}{\max(X) - \min(X)}$$

This takes in the original data and preserves relative distances between data points. So this will just be a map of the original interval (assuming inf is not a member of the original dataset otherwise its a completely different problem) onto the normalized map of $[0, 1]$.

b.) Z-score "normalization" is actually a standardization as it will only be in $[0, 1]$. For data points $x_i \in X$ rescaled into X^* is given by,

$$x_i^* = \frac{x_i - \bar{x}}{\sqrt{\sigma_X^2}}$$

Where $\bar{x}$ is the sample mean of X and σ_X is the sample standard deviation of X. This converts everything into a Z-score which is related to where the data would fall in the normal distribution.

c.) For data vector $X = [200, 300, 400, 600, 1000]$,

i.) min-max normalization with min $= 0$ and max $= 1$.

$$x_i^* = \frac{x_i - \min(X)}{\max(X) - \min(X)}$$
$$x_1^* = \frac{200 - 200}{1000 - 200} = 0$$
$$x_2^* = \frac{300 - 200}{1000 - 200} = 0.125$$
$$\vdots$$
$$x_5^* = \frac{1000 - 200}{1000 - 200} = 1$$

Giving,

$$X^* = [0, 0.125, 0.25, 0.5, 1.0]$$

ii.) z-score normalization Given X, we have $\bar{x} = 500$ and $\sigma_X \approx 316.2278$

$$x_i^* = \frac{x_i - \bar{x}}{\sqrt{\sigma_X}}$$
$$x_1^* = \frac{200 - 500}{316.2278} \approx -0.9487$$
$$x_2^* = \frac{300 - 500}{316.2278} \approx -0.6325$$
$$\vdots$$
$$x_5^* = \frac{1000 - 500}{316.2278} \approx 1.5811$$

All together, this gives,

$$X^* = [-0.9487, -0.6325, -0.3162, 0.3162, 1.5811]$$

d.) Using $X \cup \{250\}$ with the same z-score transformation, we get that x_6^* is given by,

$$x_i^* = \frac{x_i - \bar{x}_X}{\sqrt{\sigma_X}}$$
$$x_6^* = \frac{250 - 500}{316.2278} \approx -0.7906$$

Giving a final transformed X^* as

$$X^* = [-0.9487, -0.7906, -0.6325, -0.3162, 0.3162, 1.5811]$$

□

2. Question 2

Using a Chi-square test with a p-value of 0.05, determine if the following data is related or independent.

Country	Comedy	Action	Suspension	Romance	Sci-Fi
UK	120	300	250	230	100
USA	250	350	260	290	350

Ans:

To determine the Chi-square test statistic, we need to first compute the marginal distributions (makes it easier to follow at least).

Country	Comedy	Action	Suspension	Romance	Sci-Fi	Total
UK	120	300	250	230	100	1000
USA	250	350	260	290	350	1500
Total	370	650	510	520	450	2500

If we take the null hypothesis to be that genre is independent of the country, then we can calculate the expected frequency for each entry of the table using,

$$E_{i,j} = \frac{R_i C_j}{N}$$

where R_i is the row sum for the ith row, C_j is the column sum for the jth column, and N is the total number of the samples. This gives,

$$E_{1,1} = \frac{1000 \times 370}{2500} = 148$$
$$E_{1,2} = \frac{1000 \times 650}{2500} = 120$$
$$\vdots$$
$$E_{2,5} = \frac{1500 \times 450}{2500} = 270$$

This gives an expected table of,

Country	Comedy	Action	Suspension	Romance	Sci-Fi
UK	148	260	204	208	180
USA	222	390	306	312	270

The Chi-square statistic is then found by,

$$\chi^2 = \sum_{i=1}^{R}\sum_{j=1}^{C}\frac{(X_{ij} - E_{ij})^2}{E_{ij}}$$
$$\chi^2 = \frac{(120-148)^2}{148} + \frac{(300-260)^2}{260} + \cdots \frac{(350-270)^2}{270}$$
$$\chi^2 \approx 99.510$$

Here, for a dataset with 2 rows and 5 columns, the degree of freedoms is given by,

$$DoF = (R-1)(C-1) = (2-1)(5-1) = 4$$

For 4 DoF with a $\chi^2 \approx 99.510$, this is much larger than the corresponding table value for the desired p-value so we thus reject the null hypothesis and conclude that the genre preference is dependent on country.

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3. Question 3

Using the Palmer Penguins dataset, complete the following tasks.

a.) Load the dataset and report:
   i.) Number of rows
   ii.) Number of missing values per column
   iii.) Count of penguins per species

b.) Handle the missing data using a strategy (either through dropping missing data vectors or through approximating values)

c.) Explain the approach you took and justify. Ensure to identify something you'd either lose or risk with each of the following:
   i.) Dropping rows
   ii.) Imputing with global mean
   iii.) Imputing with per-species mean

d.) Standardize the numeric features. Explain why standardization is necessary for this dataset.

e.) Compute the PCA on the standardized features. Do the following:
   i.) Plot the scree plot with all possible principal components (eg percent variance captured per component).
   ii.) Compute and display the heatmap of cross-correlation matrices both before and after the PCA
   iii.) What do you notice about the off-diagonal entries of the principal components correlation matrix compared to the input feature correlation matrix, and why does this happen? Which pairs of original features were most strongly correlated?

f.) How many principal components are needed to explain 90% of the variance? Rerun PCA on the unstandardized features and compare.

g.) Plot the 2D PCA projection (PC1 vs PC2), colored by species. Comment on which species separate cleanly and which overlap. Connect this to the PCA loadings. Give an intuitive explanation of what physical features each principal components captures.

Ans:

a.) After loading the data, I get,

   i.) There are 344 rows

ii.) For the columns,

Column	Missing Values
Species	0
Island	0
Bill Length (mm)	2
Bill Depth (mm)	2
Flipper Length (mm)	2
Body mass (g)	2
Sex	11
Year	0

iii.) The penguins per species is,

Species	Count
Adelie	152
Gentoo	124
Chinstrap	68
Total	33

b.) Regarding missing values, it was handled using the global mean for numerical values and global mode for categorical values.

c.) As mentioned, I used the global mean and global mode for numerical and categorical values respectively.

i.) If you drop rows, you are throwing away data. This is fine if you have plenty to spare but this can be very costly if each data vector costs a lot to obtain (traveling to measure penguins seems like a costly venture). This is also challenging if you have very few data points without means to access more (medical trial data or for rare events for example).

ii.) The usage of the global mean can be ineffective if the per-category (species in this case) have wildly different values. Taking this in a zoo for example, if an elephant is missing its weight, it would not make sense to take the average of all animals which could include mice. This purposefully obscures data across categorical types.

iii.) The per-species mean is good but it can artificially reinforce patterns seen inside a category and lead to a reinforcement bias. For example, when I was a child, my family had an Australian Shepard that was 7-8cm taller and about 10kg heavier than any other dog of that species that I have ever seen and made the second pet of that species we got seem tiny. This is an artificial reinforcement of sampling error like what we could see if we used a per-species mean.

d.) The purpose of standardization for this dataset is that it allows for PCA to be done in a way that leaves the variance unbiased by the class or category of data.

e.) Here is the PCA data and plots.

i.) The scree plot for the Principal Components is given by,

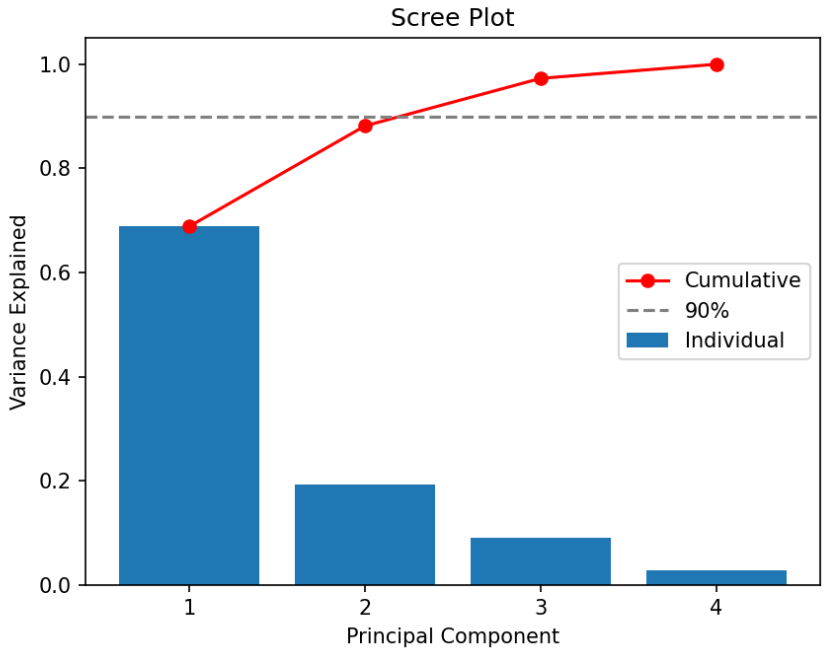


Figure 1: Scree plot of the Principal Components of the dataset.

ii.) The cross-correlation matrices before and after the PCA:

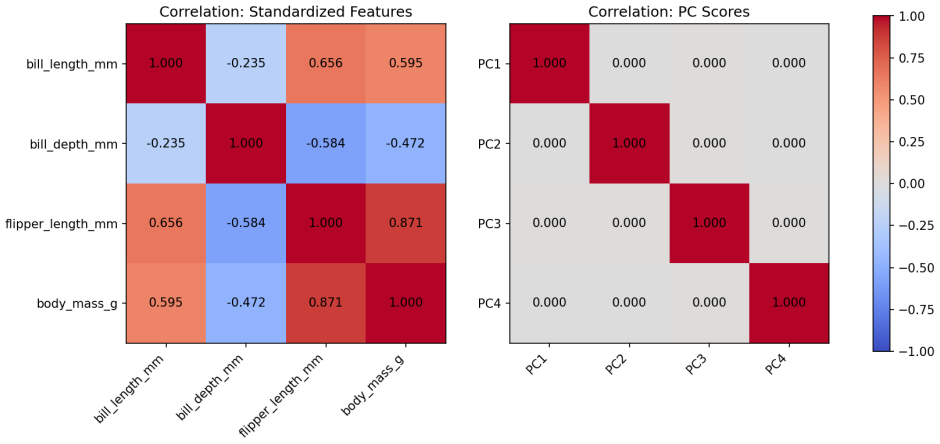


Figure 2: Cross-correlation matrix before (left) and after (right) application of PCA.

iii.) The off-diagonal components drop to zero as expected. This shows that the PCA worked since it removed variance in-line with other axis's. Before PCA, the highest absolutely correlated features were the flipper length and the body mass. For negative correlation, the flipper length and dill depth have the highest negative correlation.

f.) To explain 90% of the variance, we need 3 components (but I think an argument could be made to either use 95% of variance or to only use 2 components).

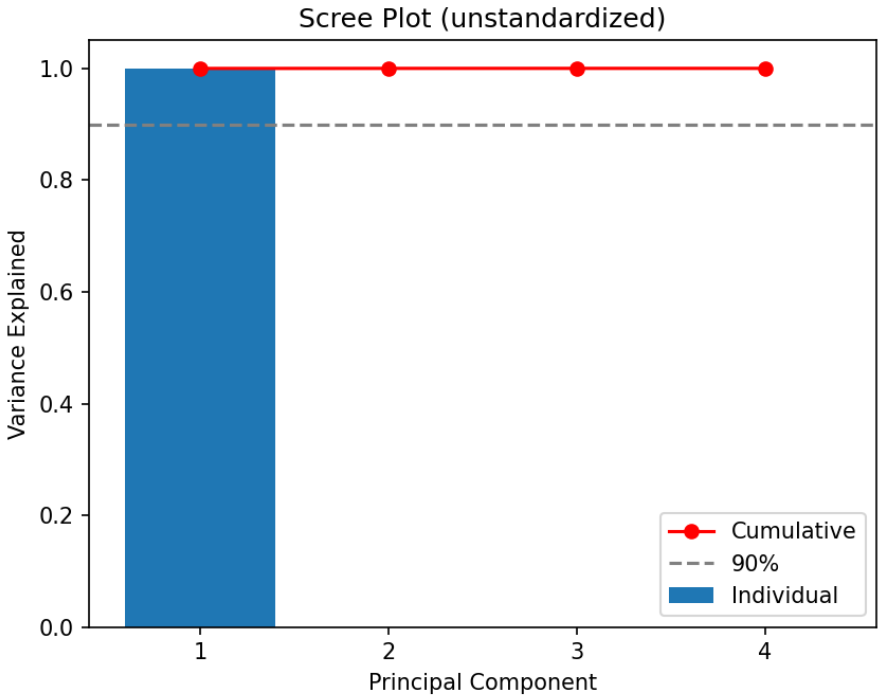


Figure 3: Scree plot of the Principal Components of the unstandardized dataset.

This is a useless plot as it tells us nothing about which data is actually useful/necessary for identification.

g.) The PCA projections.

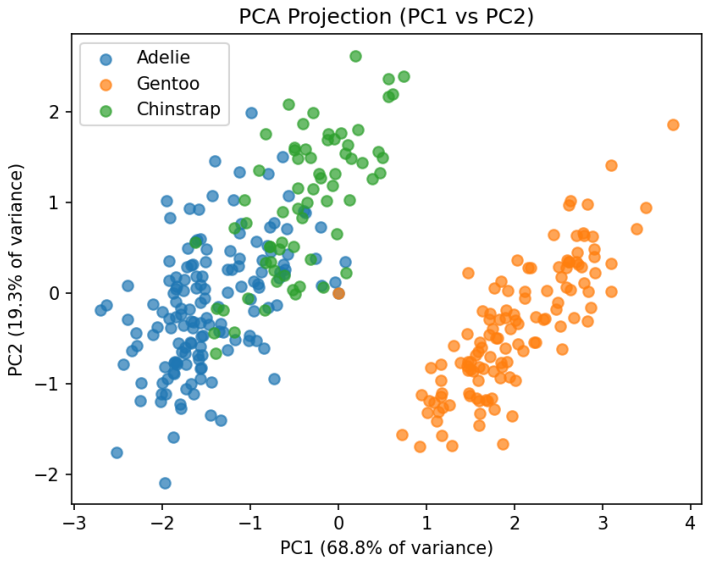


Figure 4: PCA Projection for Principal Components 1 and 2.

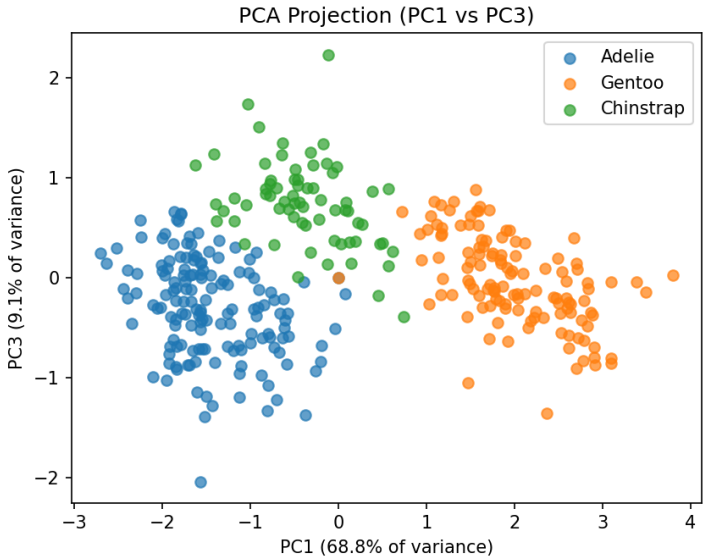


Figure 5: PCA Projection for Principal Components 1 and 3.

It should be clear that while splitting PCs 1 and 2 gives the Gentoo species its own island, I think PC1 and PC3 might be better for the identifying clusters.

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4. Question 4

Using the handwritten digits dataset, do the following:

a.) Load the data and report the shape and the number of classes. Reshape some samples into 8×8 to visualize the actual digits.

b.) Run PCA to 50 components. Plot the scree plot and cumulative explained varaiance.

c.) Visualize the 2 PCA components colored by label. Discuss what's lost compared to the full 50 dimensional representation.

d.) Reconstruct 5 random sample images from only the top k principal components for $k = 5, 20, 40$. Show the reconstructions side-by-side with the originals and discuss.

Ans:

a.) The shape for the data is $X \in [0, 16]^{1797 \times 16}$ and $Y \in \mathbb{Z}_9^{1797 \times 1}$. Since $y_i \in \mathbb{Z}_9$, there are exactly 10 classes, each one corresponding to one digits $0 - 9$. Some sample digits are shown in figure 6.

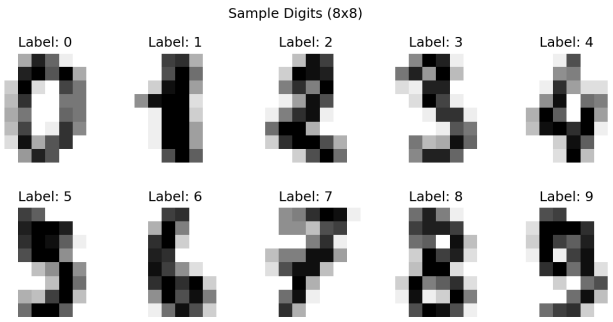


Figure 6: Representative sample of each class of the digits.

b.) Running PCA for $k = 50$, gives,

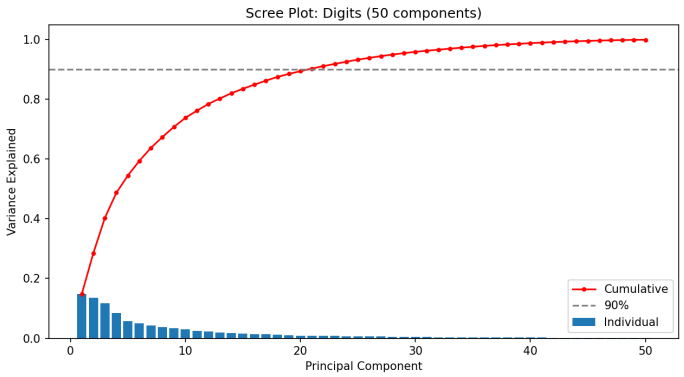


Figure 7: Scree plot and cumulative variance for the digits with $k = 50$

c.) PCA projection for PC1 and PC2

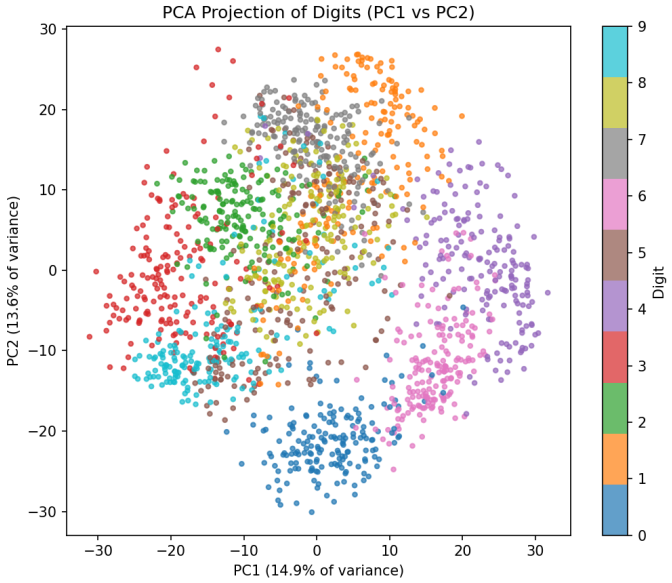


Figure 8: PCA Projection for Principal Components 1 and 2.

In doing this PCA, we lose significant structure relating to closure on the left side ($2, 3,$ and 9 are on the far left while 6 and 4 are on the right side) for PC1. For PC2, it is harder to visualize what is being lost.

d.) Reconstructing the digits gives,

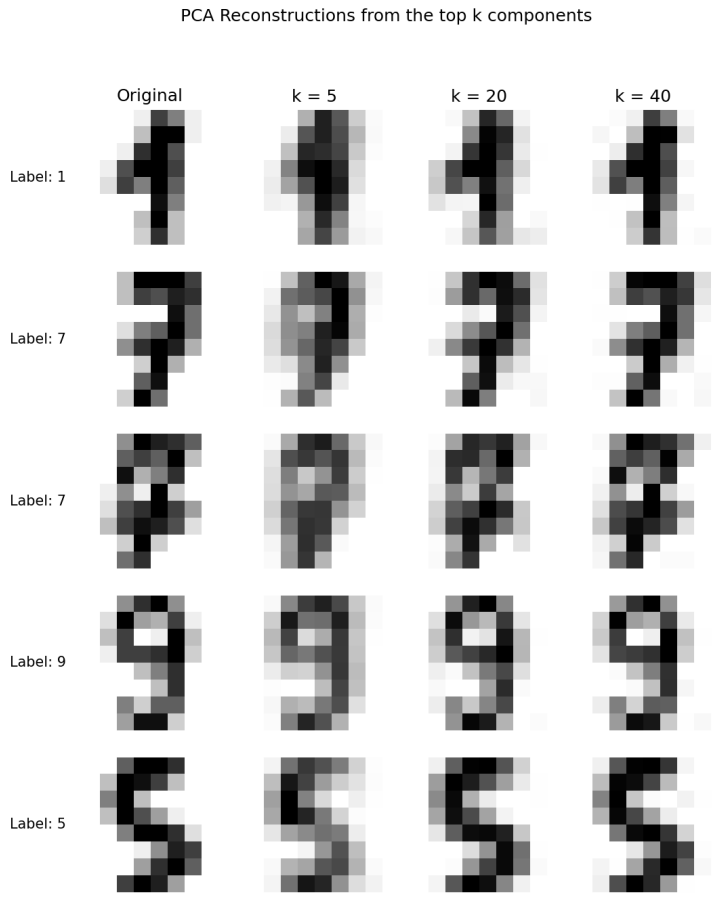


Figure 9: Reconstructions of digits using only $k = 5, 20, 40$ PC axis's.

Visually, the original vs $k = 40$ look almost identical. It is very hard to see what is changing other than more white-space being seen in the original vs the lower dimensional PCA. Even $k = 20$ still maintains a decent recognizability (for labels $\neq 7$...) while $k = 5$ loses most recognition.

□

5. Question 5

   a.) Run t-SNE on a raw high-dimensional digits dataset and separately also with the PCA-reduced (dim= 50) data to reduce to 2D data with perplexity of 5. Compare runtime and resulting embeddings.

   b.) Sweep perplexity over at least 4 values (eg $5, 30, 50, 100$). Plot the resulting embeddings in a grid and discuss cluster shape, size, and separation.

   c.) Run t-SNE twice with the same perplexity but different random seeds. Compare the two embedding. What does this tell you about interpreting cluster distance and cluster sizes in t-SNE plots.

Ans:

a.) Using the digit data again, the t-SNE is given by,

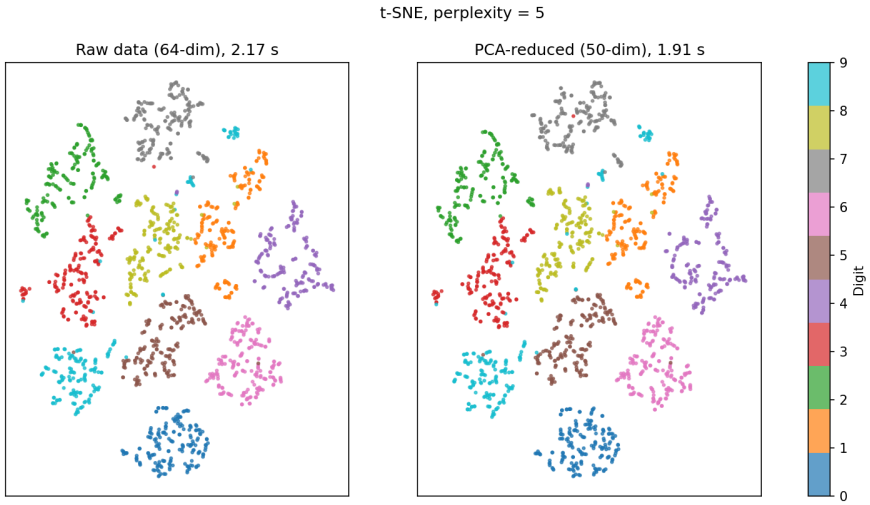


Figure 10: t-SNE vs PCA

When running the full t-SNE for non-reduced data, it is "cleaner" but runs slightly longer. Visually, the clusters are slightly tighter and not quite so spread out. The PCA is faster and honestly doesn't lose much clarity.

b.) In sweeping perplexity,

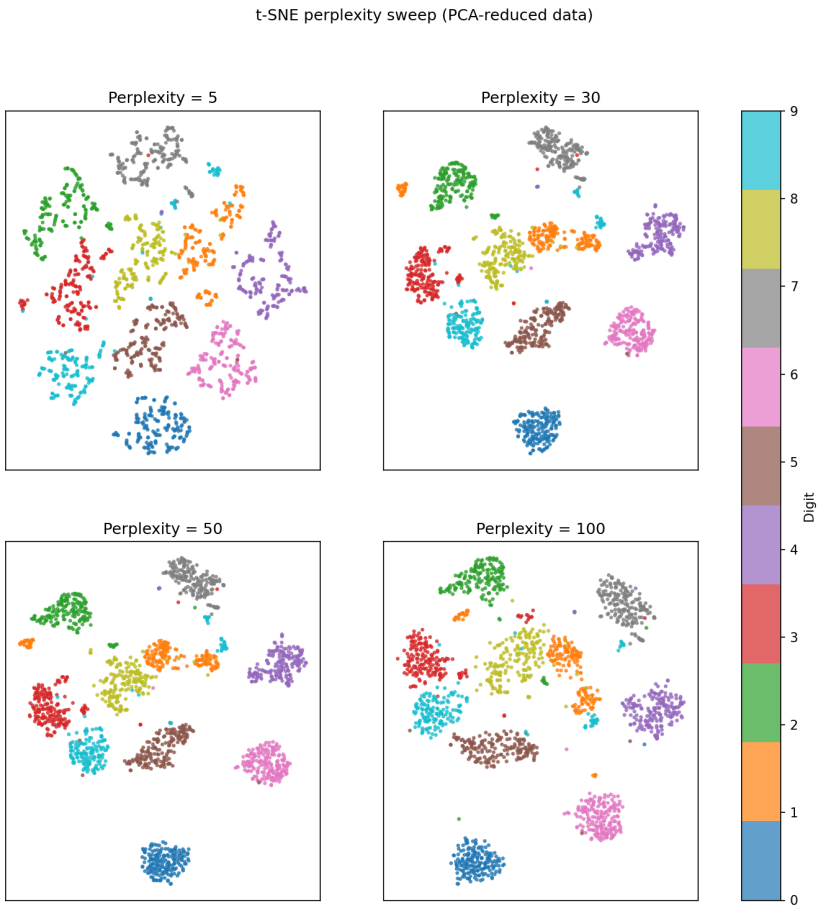


Figure 11: Sweep of potential perplexity values for t-SNE.

Running t-SNE for the varying perplexities, lower perplexity does not have a good separation but does have some clusters even if the clusters are not very tight. I would say the perplexity = 50 is the best as that has very high separation but is slightly tighter than both sweeps with 30 and 100.

c.) Running with two different seeds (137 and 29), we get the following t-SNEs,

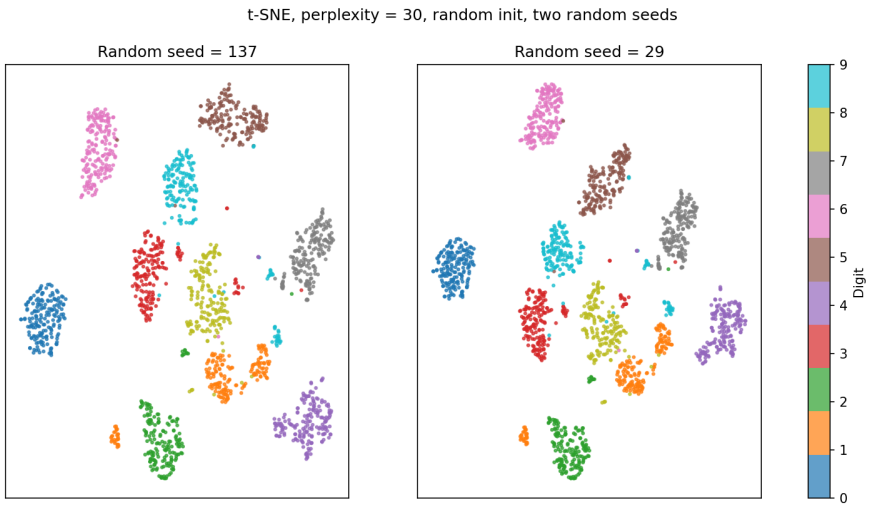


Figure 12: t-SNE with random seeds set to 137 and 29.

There are some differences for where these clusters wind up but both are sufficiently good at clustering. Realistically, I would want to do t-SNE with a variety of seeds followed by whatever post-processing I need to then ensure full statistical significance.

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6. Question 6

   Run UMAP on the same PCA-reduced digits dataset for the t-SNE to reduce into 2D, with varying number of neighbors $(5, 15, 50, 200)$ and distances $(0.0, 0.1, 0.5, 0.9)$ as a 2D grid. Discuss the effect of each parameter separately

Ans:

Running UMAP for the different values.

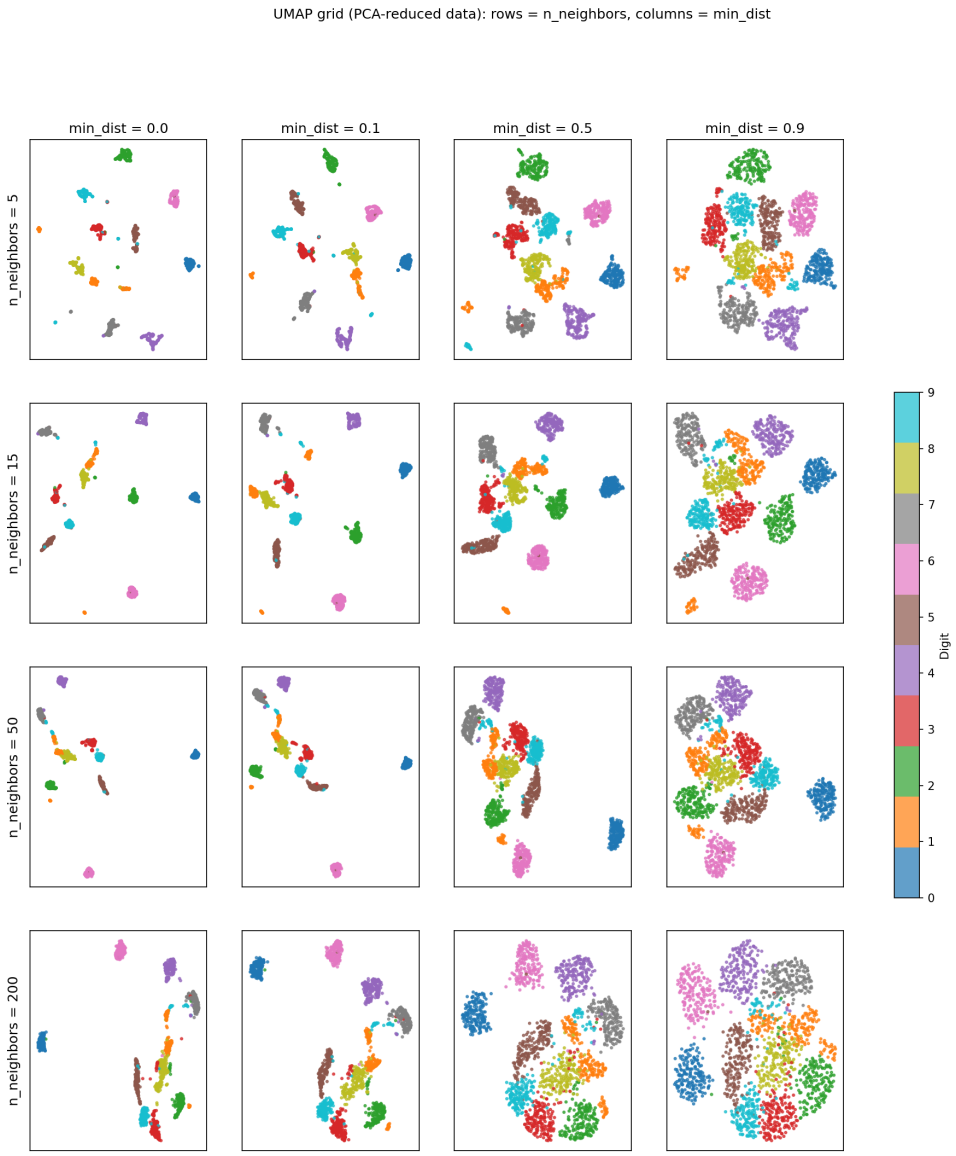


Figure 13: Grid of UMAP run with minimum distance on the X-axis from 0.0 to 0.9 and number of neighbors on the Y-axis from 5 to 200.

Increasing the minimum distance allows for overlap between clusters and does not restrict outliers as much. The number of neighbors determines how far the method will look to find similar datapoints. For example, in $N = 5$ and $d = 0.9$,

the data referring to digit 9 has multiple clusters which is could belong to. There are multiple very distinct and non-overlapping islands that push into digit 9. As we increase the number of neighbors to 200 while keeping the minimum distance at 0.9, we see that the majority of data belonging to the digit 9 moves to being in the same general region leaving the odd cluster of $1, 7, 9$ in the upper middle.

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7. Question 7

   Write a short essay explaining the following:

   a.) If you had to explain to a biologist or other experimentalist why their t-SNE or UMAP plot's cluster sizes and inter-cluster distances shouldn't be over-interpreted, how would you communicate this?

   b.) Give one concrete scenario where PCA would be a better choice than UMAP/t-SNE and one where UMAP/t-SNE would be clearly better than PCA.

   c.) Both t-SNE and UMAP are stochastic and sensitive to hyperparameters. What practical steps would you take in a real project to make sure your conclusions aren't an artifact of the random seed or hyperparameter choice?

Ans:

1. As a former experimentalist who worked in a chemical synthesis wetlab, I would communicate the lack of clear mapping from cluster measures and distances to an experimentalists by appealing to known introductory traps. My initial statement would be that its weird math and the reason they picked an experimental science and they should treat this as being a transition step in the data pipeline. For a more detailed analogy, I would use the analogy of how increasing the temperature leads to a speeding up of a chemical or enzymatic reaction. A naive interpretation of this would be if we want an enzymatic reaction to increase, we should throw it into a lava bath. Just because a trend might hold does not mean it should be blindly trusted. In a similar manner, just because we can measure a relative difference in cluster difference does not mean it is a valid methodology. Improper usage of a method without understanding both why it works and what can cause it to break will lead to bad science.

2. I would say that something which exists on the same linear manifold multiple times would benefit from PCA while anything that lives on a highly non-linear manifold that changes with each iteration would do with UMAP/t-SNE. An example might be traffic patterns which exist on the same scale each time and thus could do from repeatedly being transformed into the same PCA space. Genomic data does not live in the same space and must be redone every time (mostly) and thus would benefit from UMAP/t-SNE.

3. To account for stochasticity, I would do an ensemble sweep of the hyper parameters. So I would fit it multiple times at each hyperparameter point (or do some form of Monte Carlo sweep in practice depending on how computationally expensive of a procedure it is). This sort of sweep is infinite

dimensional (I actually am not 100% positive if the random number generator that python and numpy uses is actually random or just psuedo-random but in theory it should be random with a large number of choices up to some arbitrary machine precision) which leads this to being an intractable problem in practice. For prior passes of something like this, I would use the Monte Carlo approach but also using domain knowledge to restrict the search basins to viable options.

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